Report: Transient Grating Project

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Chapter 1

Introduction

This report summarizes the current status of transient grating signal analysis. It is intended as the companion document for the code and notebook workflow in this project.

Chapter 2

Fitting Models

For this report, three different models have been implemented and fitted to the data. The models are described in the following sections.

2.1 Model 1

The first model is described by the following equation:

$$\begin{aligned} f(t) &= \left[A_1 e^{-(t-t_0)/\tau} + A_2\right] \circledast \frac{1}{\sqrt{2\pi}\sigma} e^{-(t-t_0)^2/(2\sigma^2)} \\ &= A_1 e^{-(t-t_0)/\tau} e^{\sigma^2/(2\tau^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau}{\sqrt{2}\sigma}\right)\right) \\ &\quad + A_2 \left(1 + \operatorname{erf}\left(\frac{t - t_0}{\sqrt{2}\sigma}\right)\right) \end{aligned} \tag{2.1}$$

2.2 Model 2

The second model is described by the following equation:

$$\begin{aligned} f(t) &= \left[A_1 e^{-(t-t_0)/\tau} + A_2 + A_3 e^{-(t-t_0)/\tau_2} \sin(\omega t - \phi)\right] \circledast \frac{1}{\sqrt{2\pi}\sigma} e^{-(t-t_0)^2/(2\sigma^2)} \\ &= A_1 e^{-(t-t_0)/\tau} e^{\sigma^2/(2\tau^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau}{\sqrt{2}\sigma}\right)\right) \\ &\quad + A_2 \left(1 + \operatorname{erf}\left(\frac{t - t_0}{\sqrt{2}\sigma}\right)\right) \\ &\quad + A_3 e^{-(t-t_0)/\tau_2} e^{\sigma^2/(2\tau_2^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau_2}{\sqrt{2}\sigma}\right)\right) (\sin(\omega t - \phi) - \cos(\omega t - \phi)). \end{aligned} \tag{2.2}$$

2.3 Model 3

The third model is described by the following equation:

$$
\begin{aligned}
f(t) = &\left[A_1 e^{-(t-t_0)/\tau_1} + A_2 e^{-(t-t_0)/\tau_2} + A_3(e^{-(t-t_0)/\tau_{20}} - e^{-(t-t_0)/\tau_{10}}) + A_4\right] \sin(\omega t - \phi) \\
&\circledast \frac{1}{\sqrt{2\pi}\sigma} e^{-(t-t_0)^2/(2\sigma^2)} \\
= &\, A_1 e^{-(t-t_0)/\tau_1} e^{\sigma^2/(2\tau_1^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau_1}{\sqrt{2}\sigma}\right)\right) \\
&+ A_2 e^{-(t-t_0)/\tau_2} e^{\sigma^2/(2\tau_2^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau_2}{\sqrt{2}\sigma}\right)\right) \\
&+ A_3 e^{-(t-t_0)/\tau_{20}} e^{\sigma^2/(2\tau_{20}^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau_{20}}{\sqrt{2}\sigma}\right)\right) [\sin(\omega t - \phi) - \cos(\omega t - \phi)] \\
&- A_3 e^{-(t-t_0)/\tau_{10}} e^{\sigma^2/(2\tau_{10}^2)} \left(1 + \operatorname{erf}\left(\frac{t - t_0 - \sigma^2/\tau_{10}}{\sqrt{2}\sigma}\right)\right) [\sin(\omega t - \phi) - \cos(\omega t - \phi)] \\
&+ A_4 \left(1 + \operatorname{erf}\left(\frac{t - t_0}{\sqrt{2}\sigma}\right)\right),
\end{aligned}
\tag{2.3}
$$

where $\frac{1}{\tau_{20}} = \frac{1}{\tau_2} + \frac{1}{t_0}$ and $\frac{1}{\tau_{10}} = \frac{1}{\tau_1} + \frac{1}{t_0}$.

Chapter 3

Results

3.1 Phase Space

The available scans for the dataset CoOx12 are represented by each point in the phase space plot shown in Figure 3.1. The phase space plot is a scatter plot of the energy and intensity of the scans.

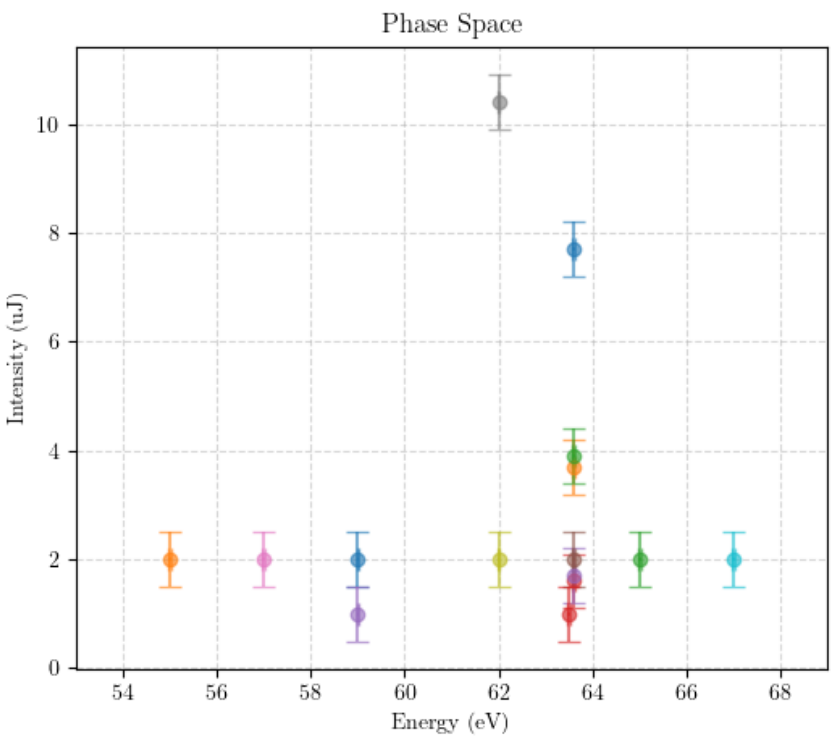

Figure 3.1: Phase space plot of the available scans for the dataset CoOx12.

3.2 Model 1

3.2.1 Constant Intensity

In this case, only scans with a constant intensity of 2 μJ are considered. Each individual signal was fitted with the Model 1 and the results are shown in Figure 3.2, together with the corresponding residuals (in green).

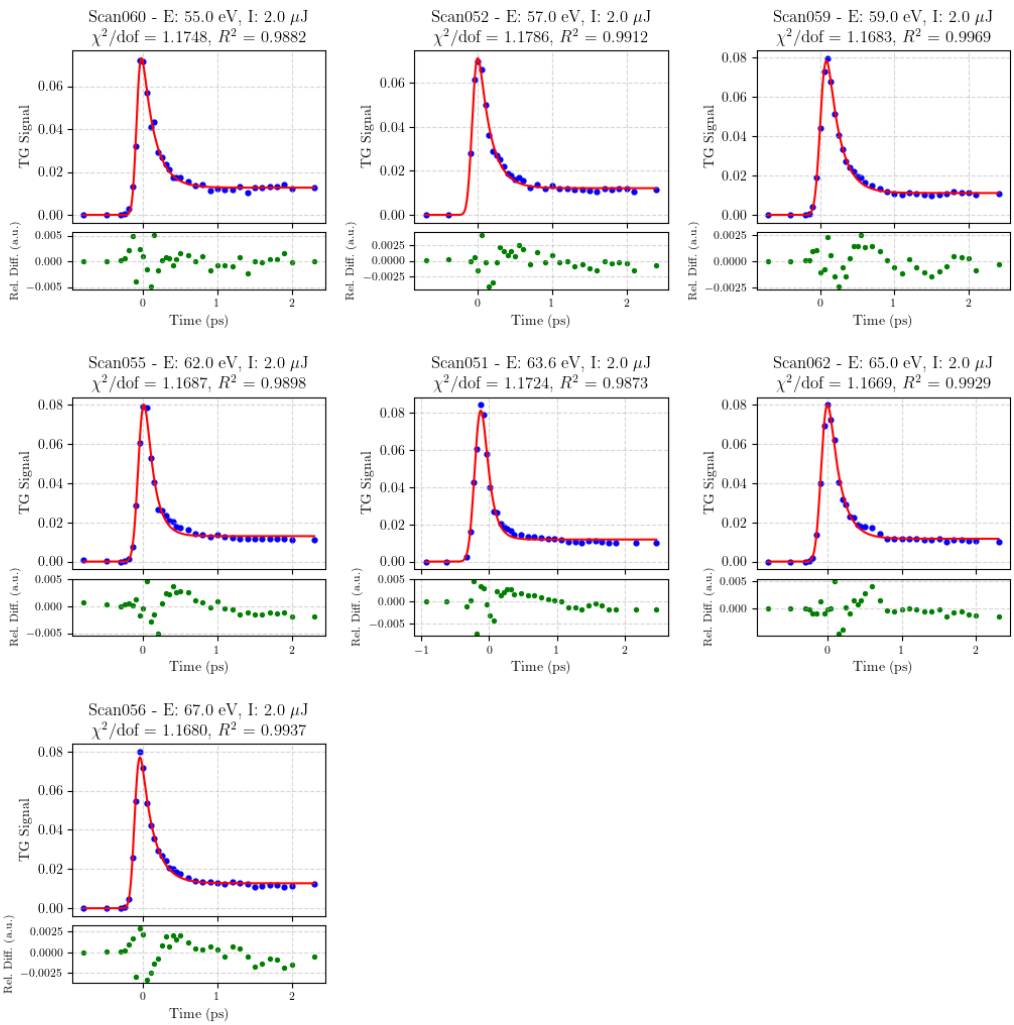

Figure 3.2: Fitted signals for the Model 1 with constant intensity of 2 μJ. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 1 are shown in Figure 3.3.

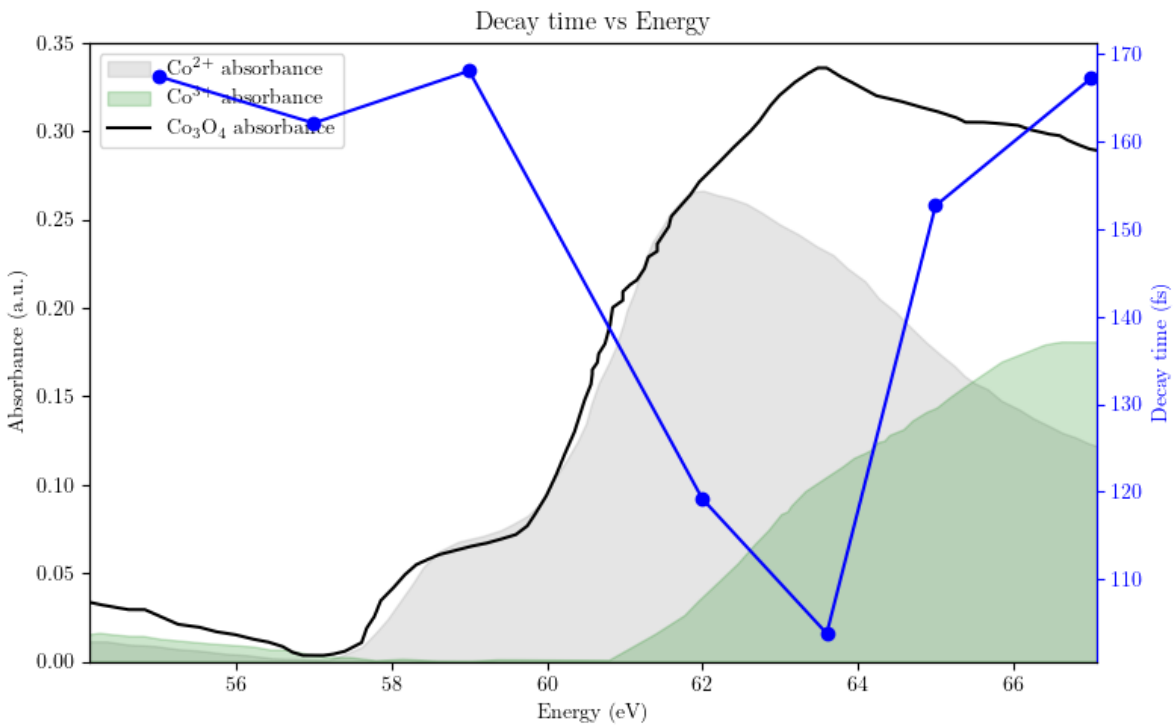

Figure 3.3: Values of τ for the Model 1 with constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. As a visual reference, values of absorbance for Co^{+2}, Co^{+3} and Co_3O_4 are shown as area plots.

3.2.2 Constant Energy

In this case, only scans with a constant energy of 63.6 eV are considered. Each individual signal was fitted with the Model 1 and the results are shown in Figure 3.4, together with the corresponding residuals (in green).

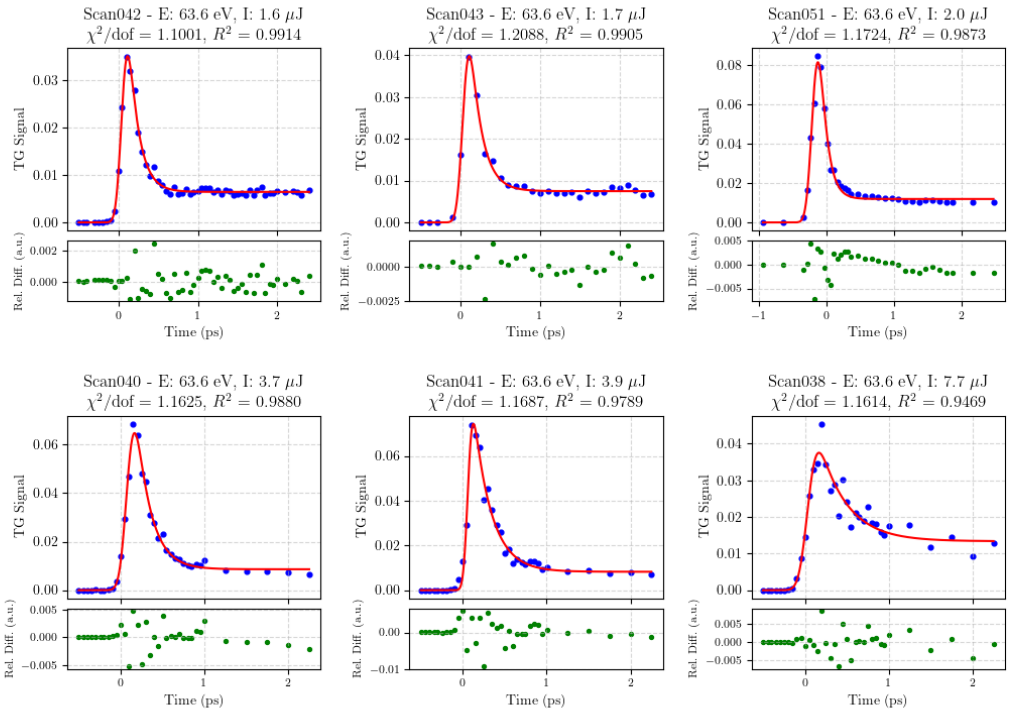

Figure 3.4: Fitted signals for the Model 1 with constant energy of 1000 eV. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 1 are shown in Figure 3.5.

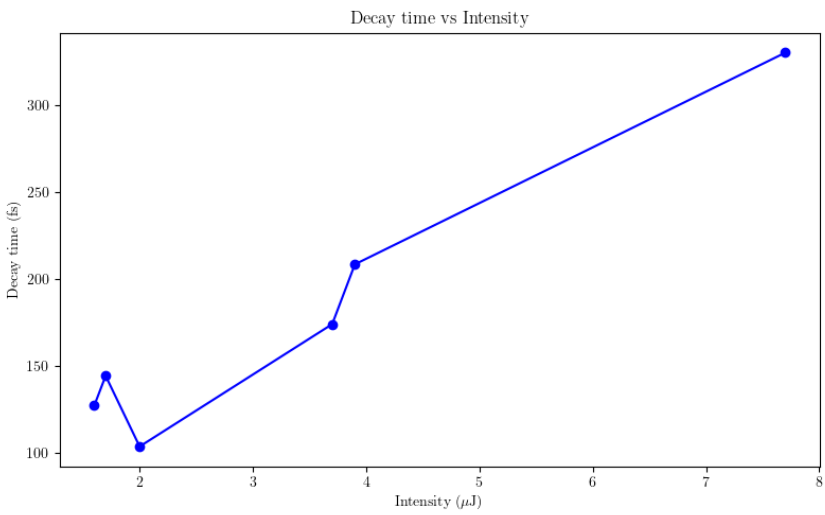

Figure 3.5: Values of τ for the Model 1 with constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

3.3 Model 2

3.3.1 Constant Intensity

In this case, only scans with a constant intensity of 2 μJ are considered. Each individual signal was fitted with the Model 2 and the results are shown in Figure 3.6, together with the corresponding residuals (in green).

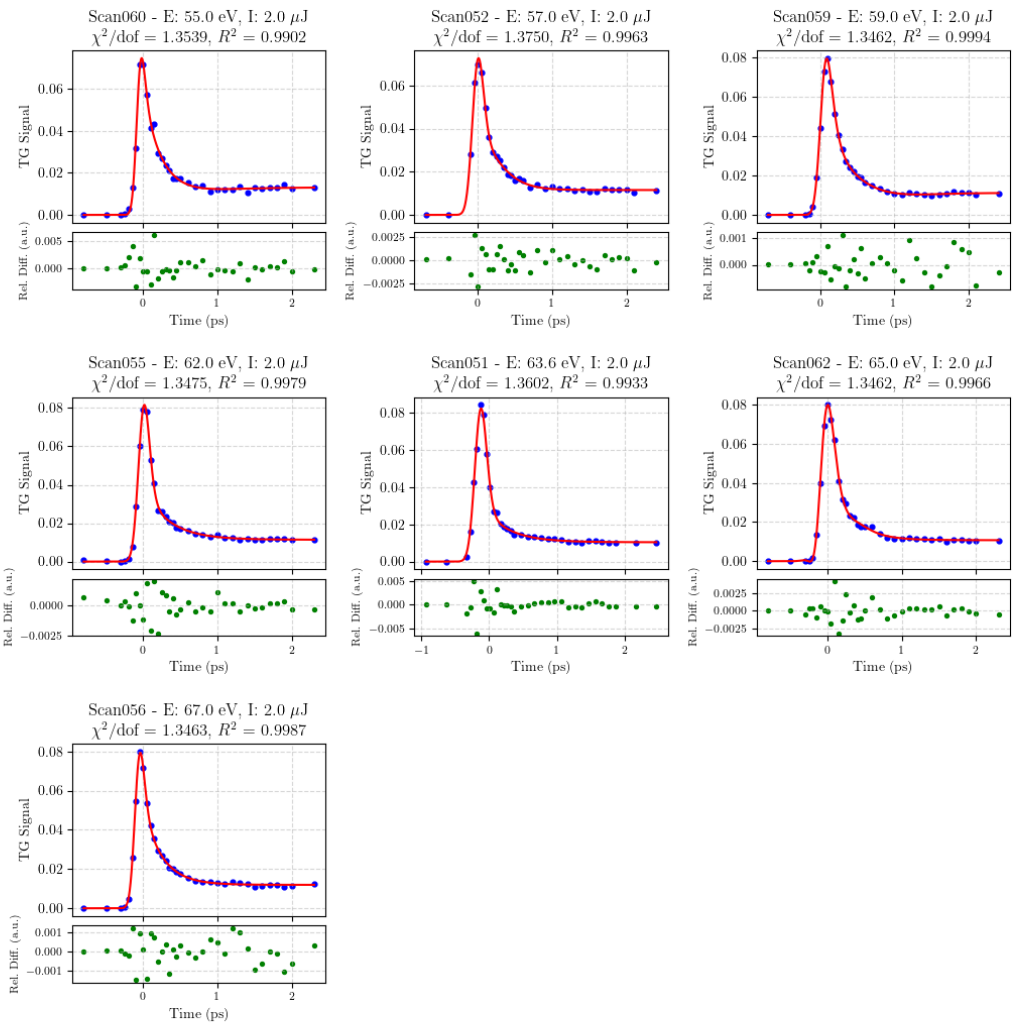

Figure 3.6: Fitted signals for the Model 2 with constant intensity of 2 μJ. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 2 are shown in Figure 3.7.

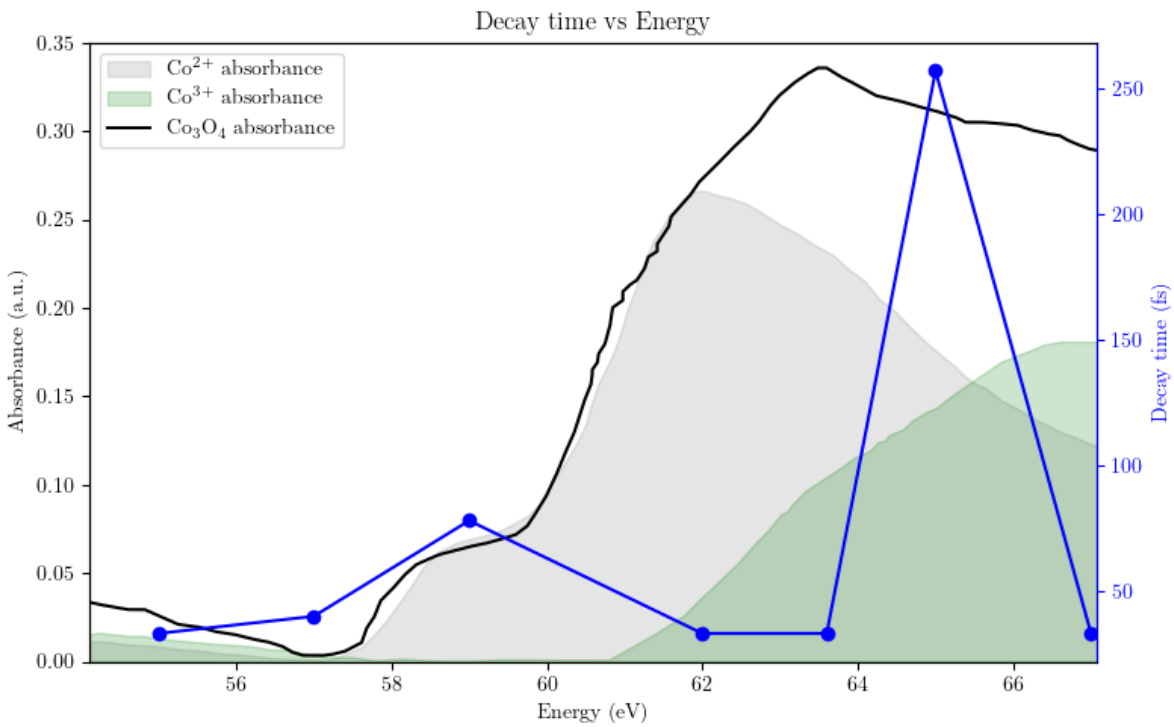

Figure 3.7: Values of τ for the Model 2 with constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. As a visual reference, values of absorbance for Co^{+2}, Co^{+3} and Co_3O_4 are shown as area plots.

The values of τ_2 for the Model 2 are shown in Figure 3.8.

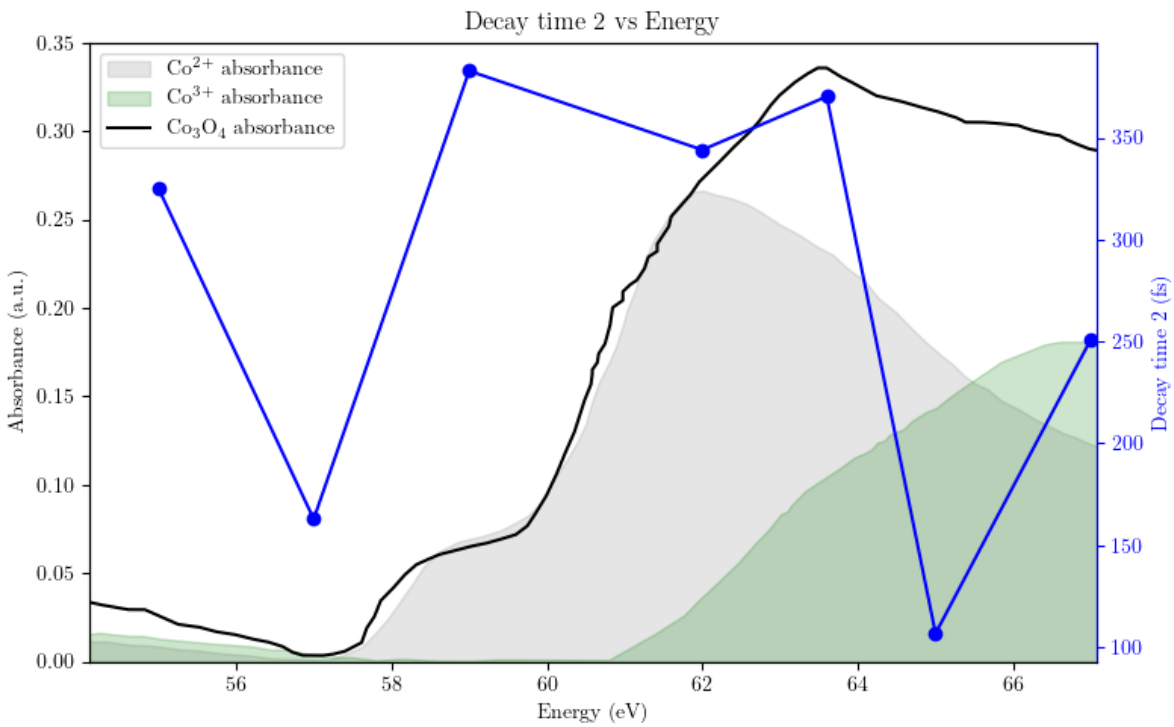

Figure 3.8: Values of τ_2 for the Model 2 with constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. As a visual reference, values of absorbance for Co^{+2}, Co^{+3} and Co_3O_4 are shown as area plots.

3.3.2 Constant Energy

In this case, only scans with a constant energy of 63.6 eV are considered. Each individual signal was fitted with the Model 2 and the results are shown in Figure 3.9, together with the corresponding residuals (in green).

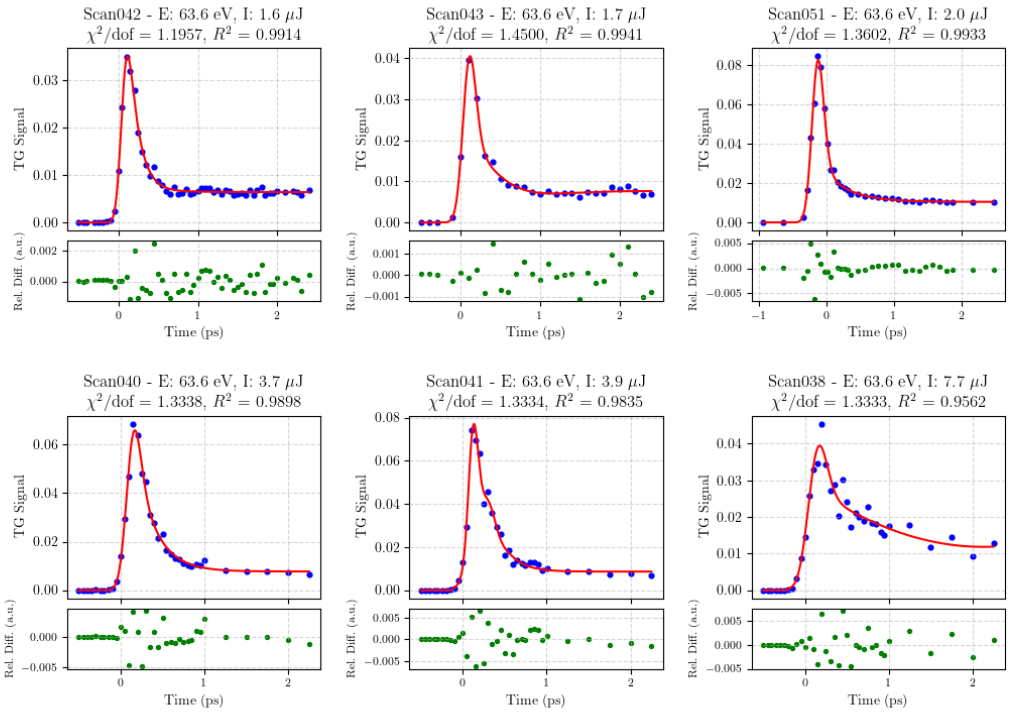

Figure 3.9: Fitted signals for the Model 2 with constant energy of 63.6 eV. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 2 are shown in Figure 3.10.

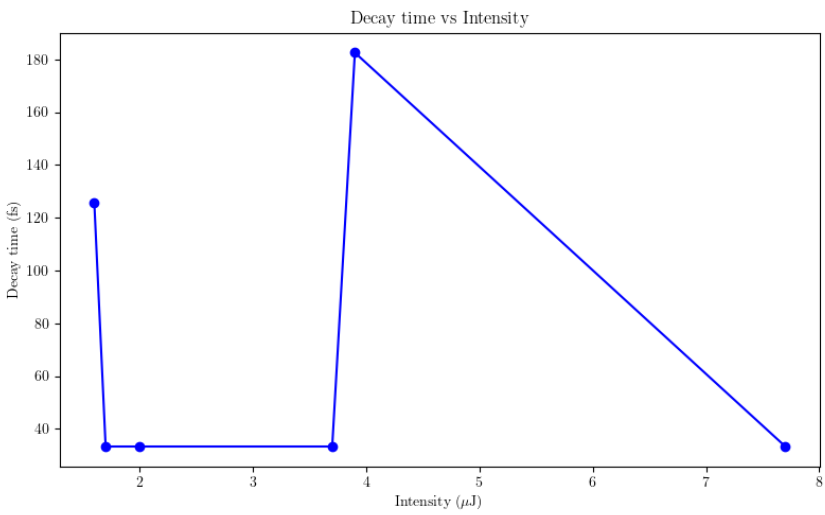

Figure 3.10: Values of τ for the Model 2 with constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

The values of τ_2 for the Model 2 are shown in Figure 3.11.

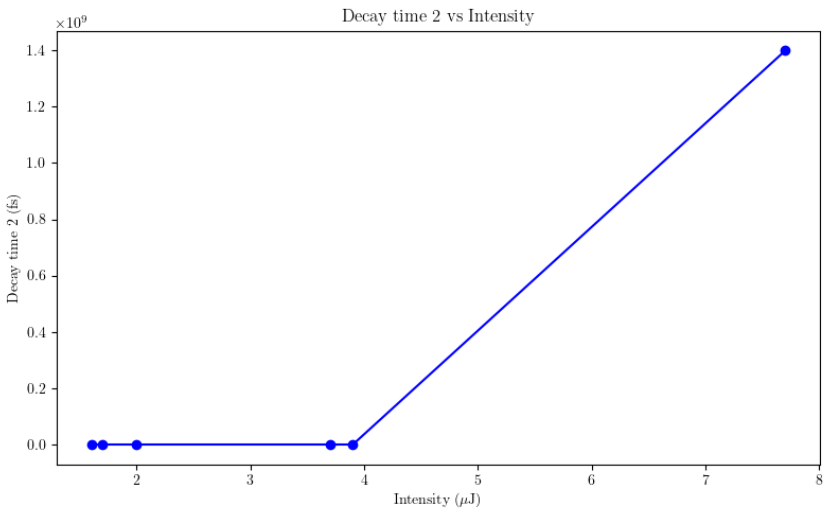

Figure 3.11: Values of τ_2 for the Model 2 with constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

3.4 Model 3

3.4.1 Constant Intensity

In this case, only scans with a constant intensity of 2 μJ are considered. Each individual signal was fitted with the Model 3 and the results are shown in Figure 3.12, together with the corresponding residuals (in green).

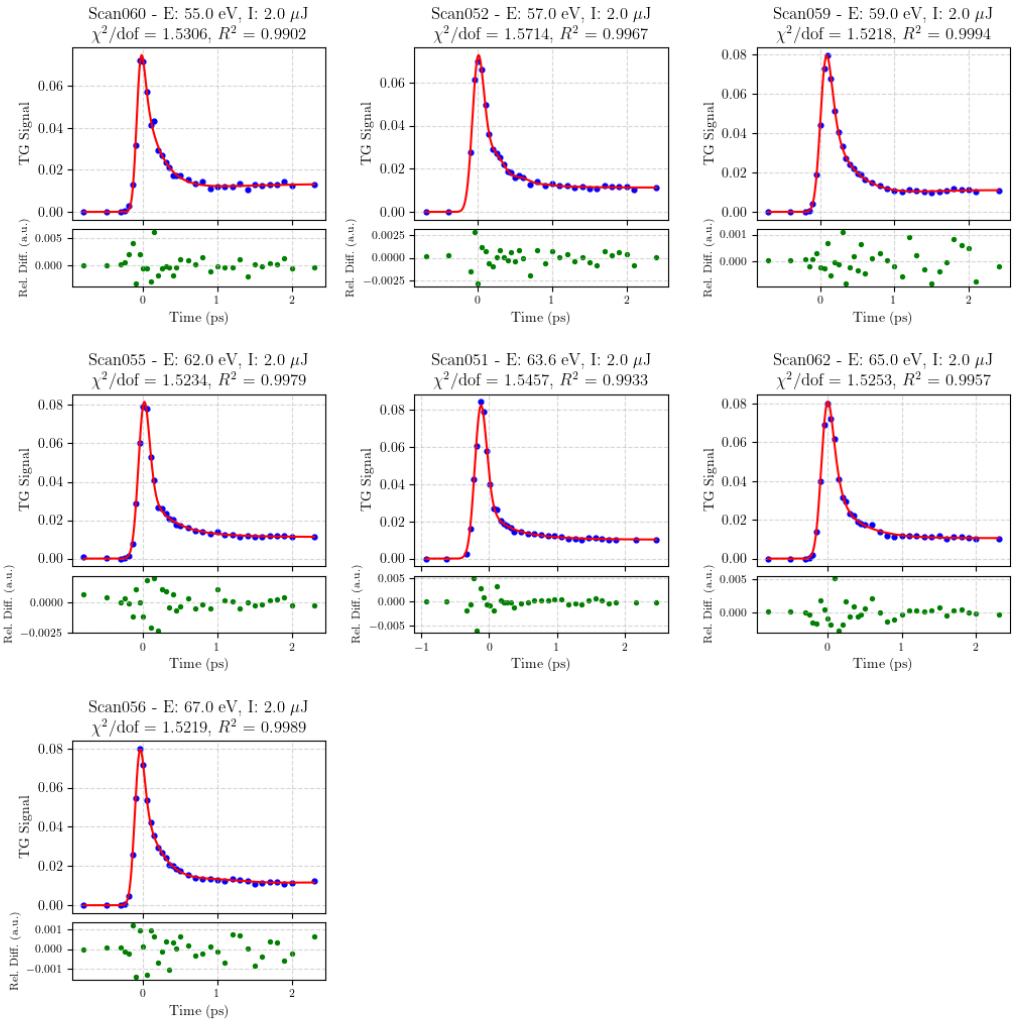

Figure 3.12: Fitted signals for the Model 3 with constant intensity of 2 μJ. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 3 are shown in Figure 3.13.

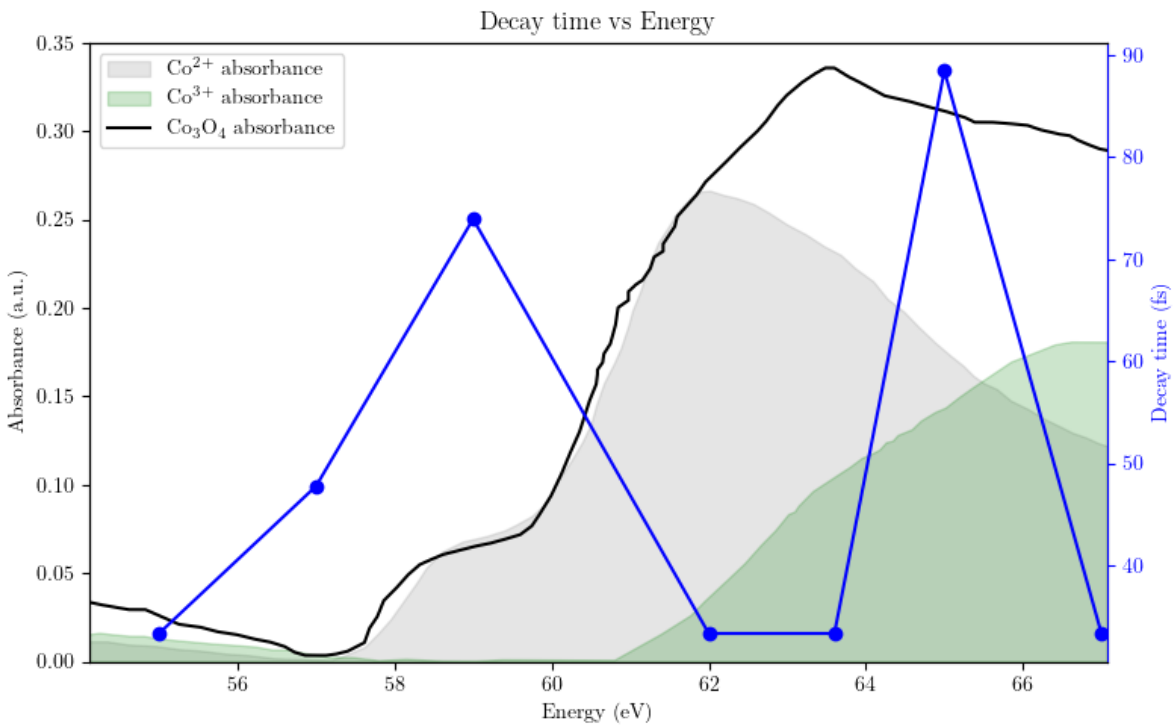

Figure 3.13: Values of τ for the Model 3 with constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. As a visual reference, values of absorbance for Co^{+2}, Co^{+3} and Co_3O_4 are shown as area plots.

The values of τ_2 for the Model 3 are shown in Figure 3.14.

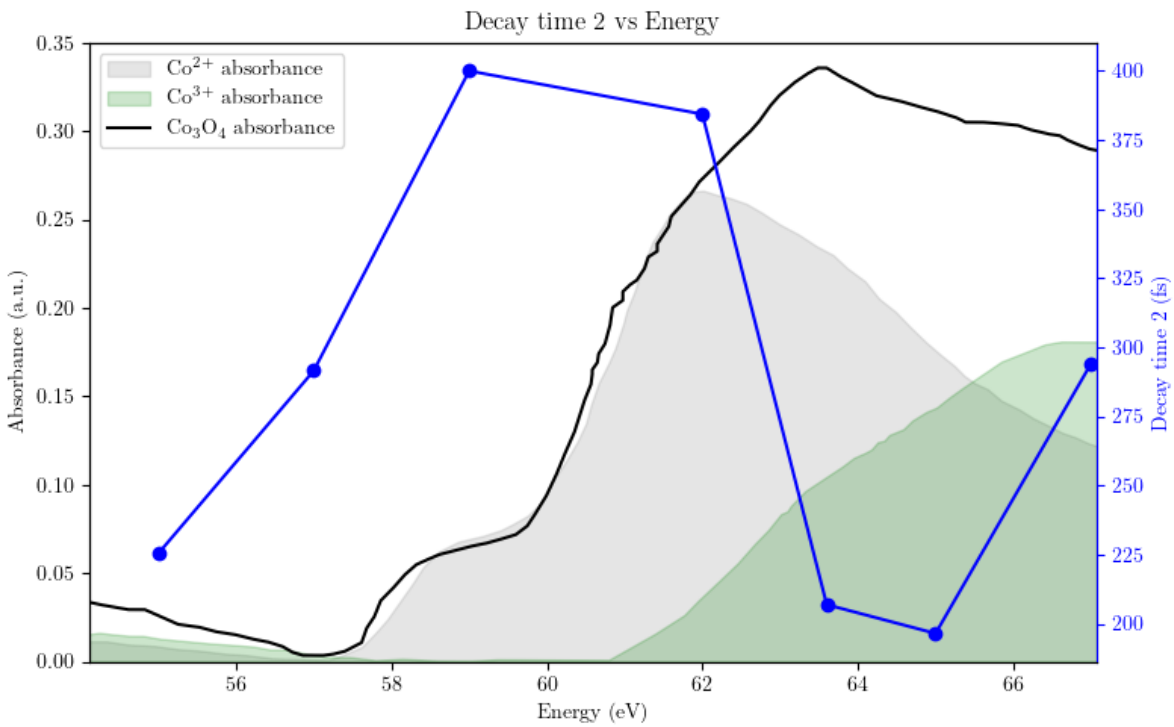

Figure 3.14: Values of τ_2 for the Model 3 with constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. As a visual reference, values of absorbance for Co^{+2}, Co^{+3} and Co_3O_4 are shown as area plots.

3.4.2 Constant Energy

In this case, only scans with a constant energy of 63.6 eV are considered. Each individual signal was fitted with the Model 3 and the results are shown in Figure 3.15, together with the corresponding residuals (in green).

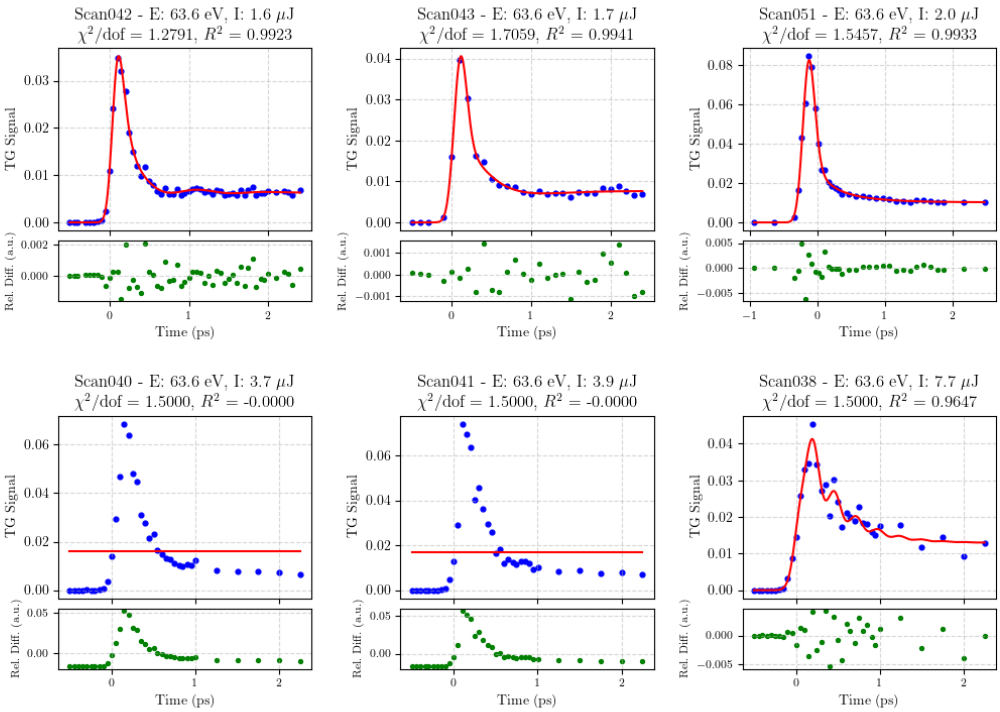

Figure 3.15: Fitted signals for the Model 3 with constant energy of 63.6 eV. The fitted curve is shown in red, the experimental data is shown in blue, and the residuals are shown in green. Values for the χ^2/dof goodness of fit are shown in the title of each plot, as well as the R^2 value.

The values of τ for the Model 3 are shown in Figure 3.16.

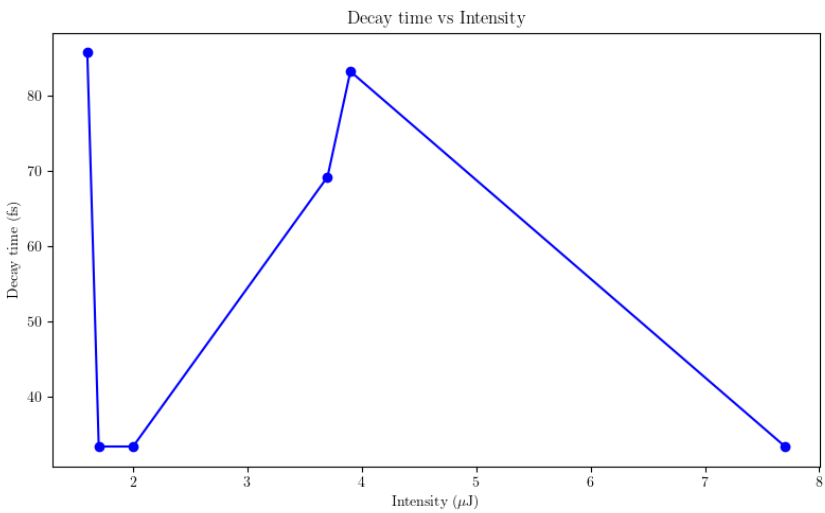

Figure 3.16: Values of τ for the Model 3 with constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

The values of τ_2 for the Model 3 are shown in Figure 3.17.

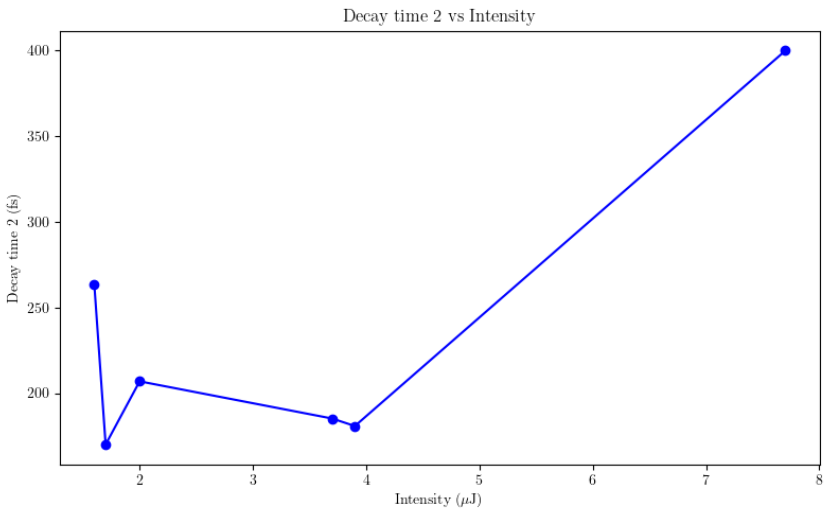

Figure 3.17: Values of τ_2 for the Model 3 with constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

3.5 Comparison of Models

3.5.1 Constant Intensity

The value of τ for the Model 1, Model 2 and Model 3 considering a constant intensity of 2 μJ are shown in Figure 3.18.

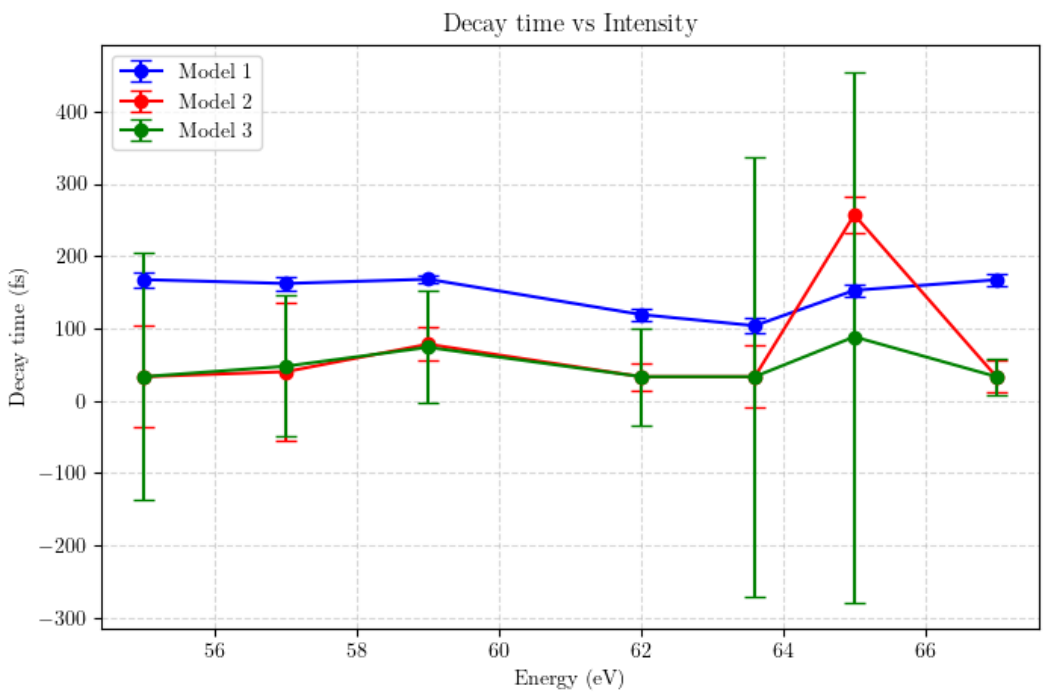

Figure 3.18: Values of τ for the Model 1, Model 2 and Model 3 considering a constant intensity of 2 μJ. The values are shown as a function of the energy of the scan.

The value of τ_2 for the Model 2 and Model 3 considering a constant intensity of 2 μJ are shown in Figure 3.19.

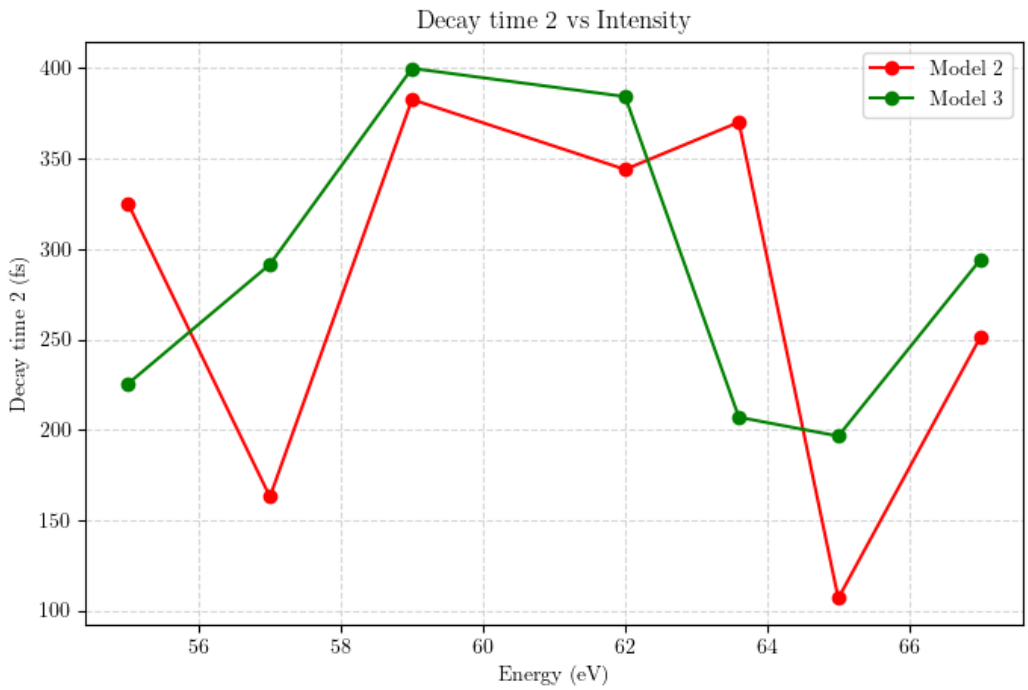

Figure 3.19: Values of τ_2 for the Model 2 and Model 3 considering a constant intensity of 2 μJ. The values are shown as a function of the energy of the scan. Errorbars were not included in the plot for clarity, since they are too big.

3.5.2 Constant Energy

The value of τ for the Model 1, Model 2 and Model 3 considering a constant energy of 63.6 eV are shown in Figure 3.20.

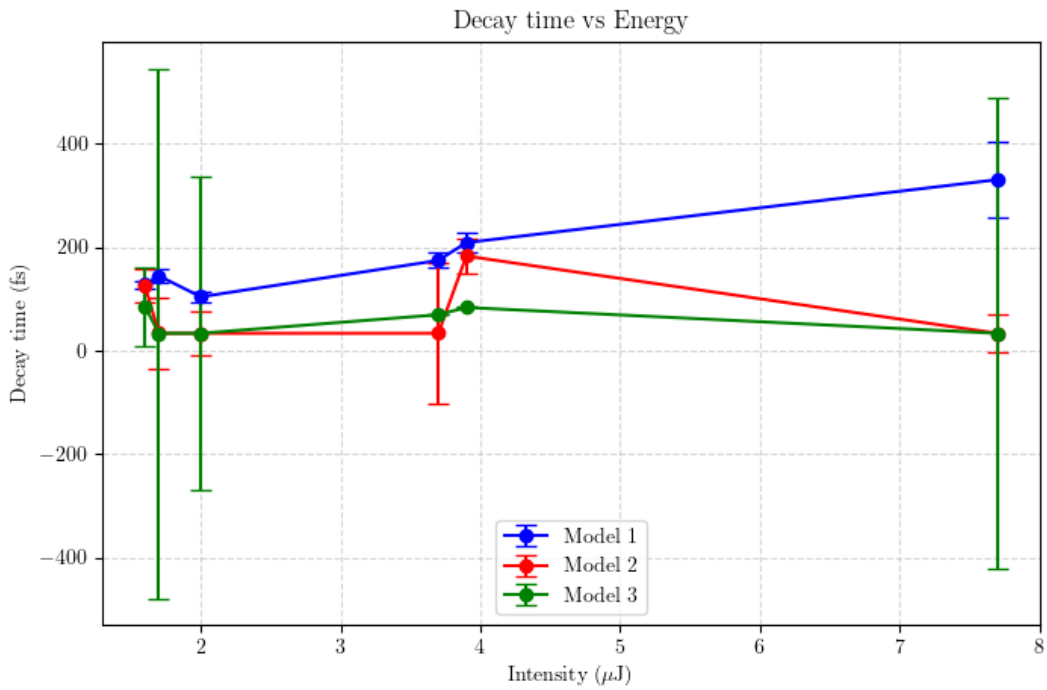

Figure 3.20: Values of τ for the Model 1, Model 2 and Model 3 considering a constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan.

The value of τ_2 for the Model 2 and Model 3 considering a constant energy of 63.6 eV are shown in Figure 3.21.

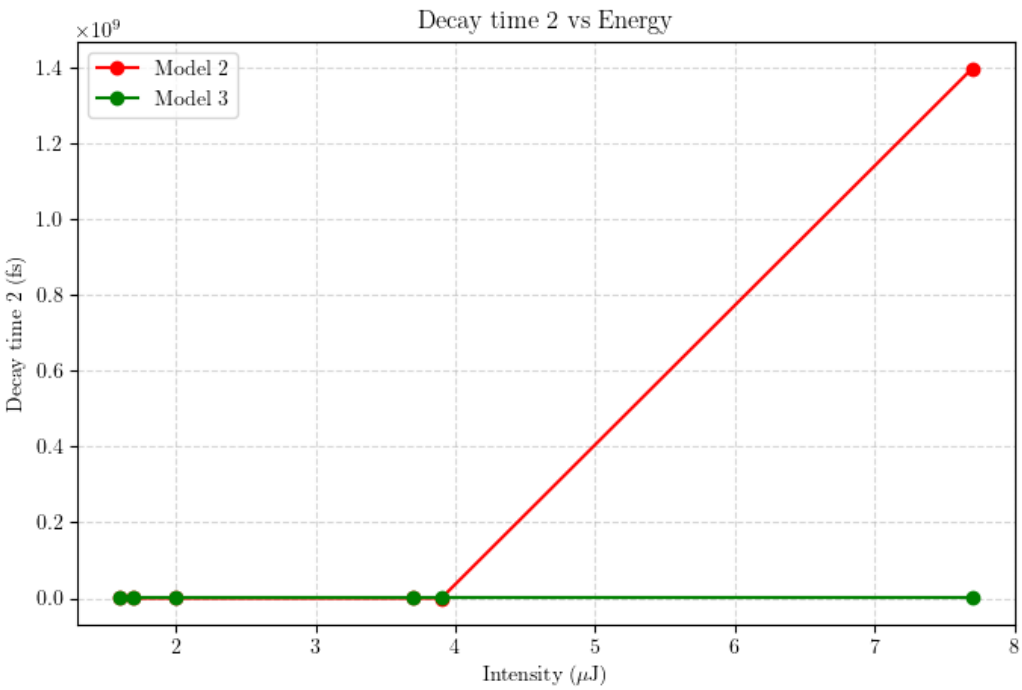

Figure 3.21: Values of τ_2 for the Model 2 and Model 3 considering a constant energy of 63.6 eV. The values are shown as a function of the intensity of the scan. Errorbars were not included in the plot for clarity, since they are too big.

3.6 Observations

3.6.1 Constant Intensity Results

It is observed that all models can fit the data well. However, Model 1 leads to a χ^2/dof goodness of fit closer to 1 than Model 2 and Model 3. However, Model 2 and Model 3 can fit the data better than Model 1 since the value of R^2 is closer to 1 for Model 2 and Model 3. Also, by observing the residuals, it is observed that Model 2 and Model 3 are able to fit the data better than Model 1, since the residuals are smaller for Model 2 and Model 3.

3.6.2 Constant Energy Results

Similar to the constant intensity results, all models can fit the data well. However, Model 1 leads to a χ^2/dof goodness of fit closer to 1 than Model 2 and Model 3. However, Model 2 and Model 3 (except for intensity values of 3.7 μJ and 3.9 μJ) can fit the data better than Model 1 since the value of R^2 is closer to 1 for Model 2 and Model 3. Also, by observing the residuals, it is observed that Model 2 and Model 3 are able to fit the data better than Model 1, since the residuals are smaller for Model 2 and Model 3.

Nevertheless, Models 2 and 3 lead to τ vs intensity plots that do not show a clear trend, while Model 1 does.

3.6.3 Comparison of Models Results

At a constant intensity of 2 μJ, Models 1 to 3 show τ vs energy plots that show a similar trend. However, the values of τ for Models 2 and 3 are consistently lower than the values of τ for Model 1. Also, for Models 2 and 3, error bars are noticeably bigger than the ones obtained for Model 1.

At a constant energy of 63.6 eV, only Model 1 shows a clear trend in the τ vs intensity plot while keeping error bars small. Models 2 and 3 lead to τ vs intensity plots that do not show a clear trend and keep error bars big.